Practical 2- Binary to Gray code conversion

Aim:- To verify & realize Binary to Gray code conversion using XOR gate Trainer Kit

Toolkit Used:-

1) DB06 Binary to Gray code converter.
2) Digital trainer kit with input power supply.
3) +5V DC Power Supply Adapter.
4) Patch cords.

Theory:-

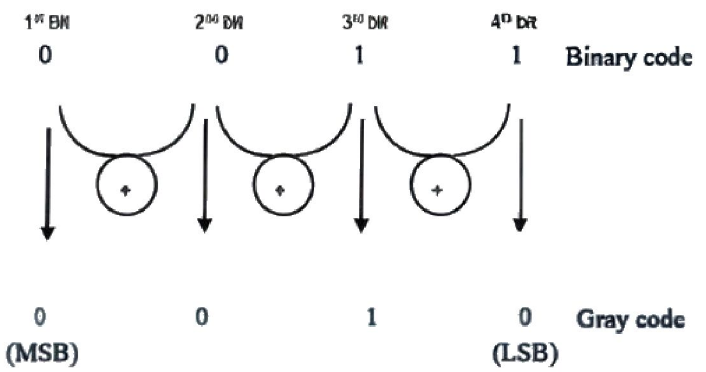


Fig. 1 Binary to Gray Conversion

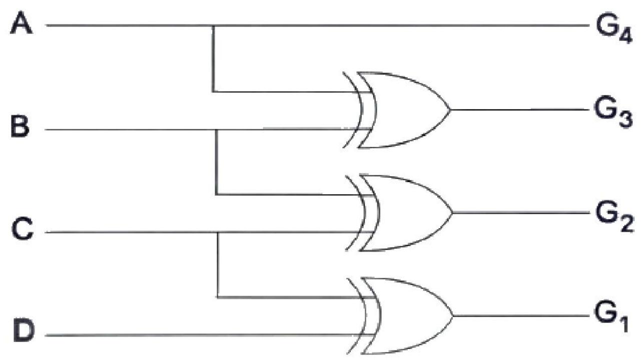


Fig. 2. Binary to Gray Code Conversion Circuit Diagram

For Binary to Gray code conversion using XOR gates from figure 1,

1. The MSB of the Gray code will be **0,** same as of the MSB of Binary code is **0**.
2. Now, the second gray bit will be XOR of 1st & 2nd binary bits as, **0 ⊕ 0= 0.**
3. Similarly, third gray bit will be XOR of 2nd & 3rd binary bits as, **0 ⊕ 1= 1**.
4. And Lastly, LSB gray bit will be XOR of 3rd & 4th binary bits as, **1 ⊕ 1= 0**
5. Therefore the equivalent Gray Code for the given binary code 0011 will be **0010**.

Observation Truth Tables:-

XOR

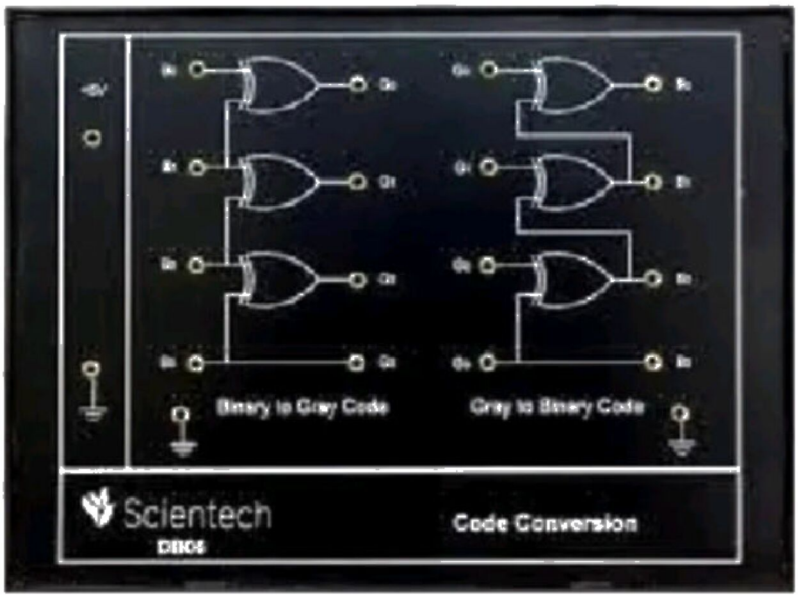
A	B	Output
0	0	0
1	0	1
0	1	1
1	1	0

Table 1: Truth Table of Binary to Gray Code Conversion

Binary Code				Gray Code			
B3	B2	B1	B0	G3	G2	G1	G0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

Connections:-

1. Connect +5 V pin from the trainer on the left side to +5V pin on the DB06 using a patch cord wire.
2. Connect GND pin from the trainer to ground symbol pin on the DB06.
3. Connect the input section pins no. D3, D2, D1 and D0 on the bottom of the trainer to B0, B1, B2, and B3 pins of the DB06 binary to gray code circuit on the left of the panel respectively.
4. Connect the pin no. G0, G1, G2, and G3 of DB06 binary to gray code circuit to the output pin no. D3, D2, D1 and D0 of the 8-bit LED Display of the trainer respectively, in order to display the gray code from the binary inputs.
5. Make sure all your connections are right then turn on the power supply.
6. Turn D3 switch to position 1 and keep D0, D1 and D2 switches at position 0 then observe the output on 8-bit LED display (green light indicates 0 and red light indicates 1).
7. Observe the output for different input combination as shown in truth table of binary to gray code conversion on Table (1).
8. Verify Truth Table on Table (1).



Results: Binary to Gray code conversion has been verified & realized.